Which Metadata Can the PE Vote See?

Experiment 1 results and a re-annotation plan

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Two scores per attribute: what the vote can use vs. what ϕ encodes

ϕ = sentence-t5-base, the embedding of MAPLE's L_2 1-NN vote. bioRxiv, 9 Gemini-labelled attributes.

k-NN visibility. For each point, the share p_k of its 5 nearest neighbours under ϕ with the same value of attribute k; chance level $p_k^0 = \sum_v P_k(v)^2$:

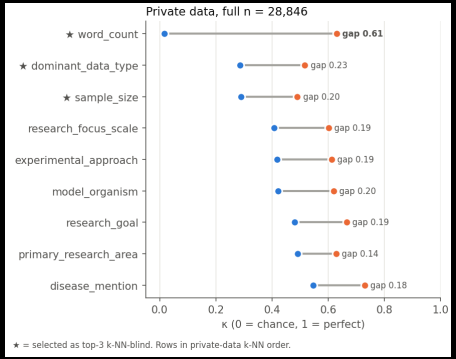
$$\kappa_k^{\mathrm{NN}} = \frac{p_k - p_k^0}{1 - p_k^0}.$$

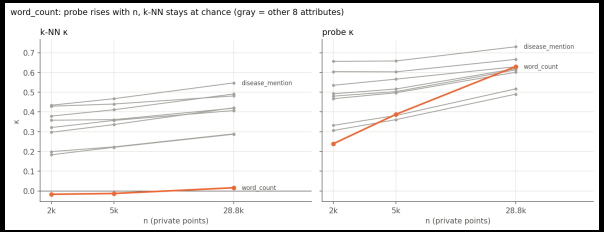
Probe visibility. Logistic regression on standardized ϕ ($C = 1$), scored by Cohen's $\kappa_k^{\mathrm{probe}}$.

Both on the same random half/half splits (20 repeats; 5 at full n), at $n = 2\mathrm{k}, 5\mathrm{k}, 28{,}846$. A large **gap** $\kappa_k^{\mathrm{probe}} - \kappa_k^{\mathrm{NN}}$ means: k is in ϕ, but L_2 neighbourhoods are dominated by other directions. The 3 attributes with lowest κ^{NN} are the quota candidates.

Toy check: `absent` $0/0$, `quiet` $0/0.99$, `loud` $0.94/0.99$ (k-NN / probe).

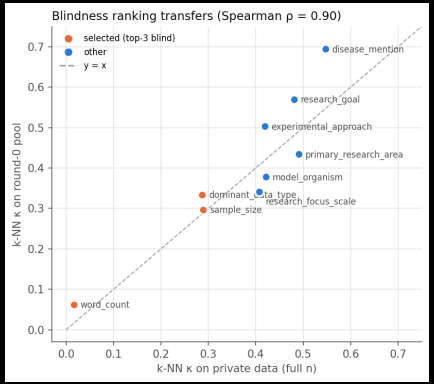
Private data: length is decodable from ϕ but invisible to 5-NN





- Left: blue κ^{NN}, orange κ^{probe}, full n.
- **word_count**: probe $0.24 \to 0.39 \to 0.63$ with n; k-NN stays ≈ 0. Gap 0.61, $3\times$ any other.
- dominant_data_type, sample_size: weakly encoded (probe ≈ 0.5, k-NN ≈ 0.29). Other 6: gap 0.14–0.23.
- Top 3 identical at all three n; sd ≤ 0.026.

The blindness ranking transfers to the round-0 pool, so selection needs no private data



Pool. Qwen2.5-7B, MAPLE's prompt, AIM $\varepsilon = 4$ rows; $14{,}000$ texts saved *before* the vote. Labels = requested AIM values; word_count measured from the text.

Transfer. Spearman $\rho = 0.90$ ($p = 0.001$) between private and pool κ^{NN}; same top 3 and same bottom. On the pool, κ^{NN} is **post-processing** of AIM output.

Length compliance is poor. Probe κ: requested length 0.22, measured length 0.70; 10 length bins in the pool vs. 12 requested. A length quota must use measured length.

Most content attributes are *more* visible in the pool (disease_mention 0.69 vs. 0.55), likely because the prompt asks for every value explicitly. Same-prompt 1-NN rate is 1.3%, so this is not a prompt-twin effect.

Experiment 1b: correction measures how much of the drift the real vote removes

Run MAPLE's vote on the round-0 pool: each private point votes for its nearest pool text, Gaussian noise, keep top $K = 2000$. For each attribute, compare three label distributions against the private target:

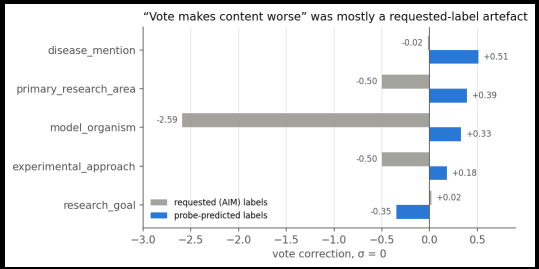
$$\text{correction}_k = \frac{\text{JSD}_{\text{random}} - \text{JSD}_{\text{vote}}}{\text{JSD}_{\text{random}} - \text{JSD}_{\text{floor}}}.$$

- $\text{JSD}_{\text{random}}$: K pool texts drawn uniformly (the generator's drift).
- $\text{JSD}_{\text{floor}}$: K private texts resampled (finite-sample noise).
- 1 = vote fully removes the drift, 0 = no better than random, < 0 = vote adds drift.

Noise $\sigma \in \{0, 2.58, 4.75, 6.87\}$, i.e. $\varepsilon = \infty, 4, 2, 1$ in MAPLE's schedule.

Which labels for the pool texts? Requested AIM values, or labels predicted from the text by a probe trained on private Gemini labels.

Requested labels hide generator drift and make the vote look harmful

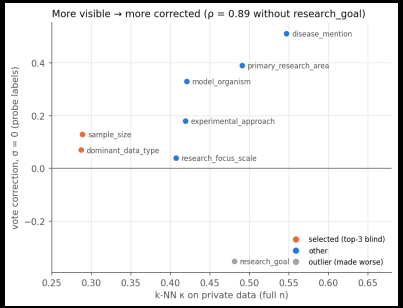


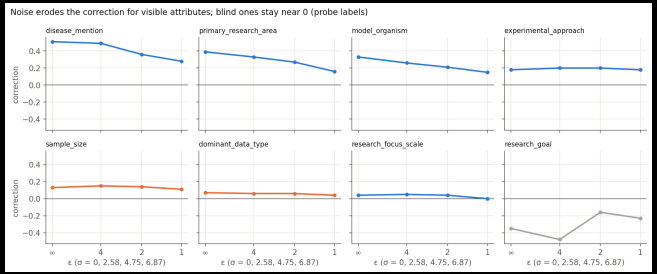
With requested labels the vote seemed to damage content: model_organism -2.59, primary_research_area -0.50.

The vote selects texts by what they **say**, but requested labels record what was *asked*. The LLM follows requests only partly (compliance κ 0.47–0.76), so selected texts carry skewed requested labels even when their content improved. word_count_requested shows the extreme case: -12.4 while measured length improves by $+0.50$.

Requested labels also match the AIM marginals by construction, so they hide the generator's drift: on probe labels, $\mathrm{JSD}_{\mathrm{random}} = 0.07$–$0.17$ vs. floor ≈ 0.02–0.04.

The vote corrects visible attributes; blind ones stay 4–7× above the floor





- Correction at $\sigma = 0$: disease_mention 0.51, primary_research_area 0.39, model_organism 0.33; sample_size 0.13, dominant_data_type 0.07. Private κ^{NN} predicts it: $\rho = 0.89$ ($p = 0.007$, 7 attributes). Outlier **research_goal**: -0.35 at probe κ 0.67.
- Noise erodes correction of visible attributes (disease_mention $0.51 \to 0.28$ at $\varepsilon = 1$); blind ones stay near 0.

Re-annotation: label once with an LLM, then classify every round cheaply

Probe labels are circular: built on ϕ, they see each attribute as the vote does. Testing whether the generator follows metadata needs fresh, ϕ-independent labels in every PE round.

Pipeline (`reannotate/`, set up 2026-10-11).

1. Qwen2.5-7B annotates the $14{,}000$-text pool and $3{,}000$ private texts: MAPLE's prompt, guided JSON over the schema, greedy.
2. Per-attribute classifiers trained on Qwen pool labels (70/15/15 split by prompt group): TF-IDF + LR; fine-tuned SciBERT, BiomedBERT; 2-layer MLPs (per-attribute or shared trunk + heads) on frozen ϕ or frozen BiomedBERT.
3. Evaluate on pool test, private vs. Qwen, private vs. Gemini: Cohen's κ, macro-F1, marginal JSD (raw and confusion-corrected).

Role split. ϕ-MLPs cost almost nothing, since the vote computes ϕ anyway: candidate **quota annotator**, never the drift evaluator. The per-attribute κ gap (best text encoder − best ϕ model) shows whether an attribute's information is in ϕ at all.

Caveats and next steps

Caveats

- Length bins are ordinal but scored as nominal; both κ values understate how much length ϕ sees. In 1b the vote still closes half the length gap (JSD $0.42 \to 0.23$, floor 0.03).
- The two chance corrections differ ($\sum p^2$ vs. Cohen), so gaps compare only roughly across attributes.
- Qwen annotates its own generations; Qwen-vs-Gemini agreement on private texts bounds this.
- Experiment 1 on private data is a non-private diagnostic; the method uses only the pool.

Next

- Reannotate results (pending); Gemini labels for the pool once the API key arrives.
- Length compliance: requested vs. measured (MAE, histograms); length-regression probe.
- Experiment 2: quota on measured word_count, sample_size, dominant_data_type; JSD and MAUVE vs. MAPLE across ε.

Takeaway

Length is encoded in ϕ but ignored by its L_2 neighbourhoods; dominant_data_type and sample_size are weakly encoded. The same three attributes are selected from the privacy-free round-0 pool.

Scored on text-derived labels, the vote corrects attributes in proportion to their k-NN visibility and leaves the blind ones uncorrected, more so under noise: the target of the quota.

The re-annotation pipeline supplies the per-round, ϕ-independent labels Experiment 2 needs.